

Seminar 3

1. From

$$\text{VaR}_\alpha(L) = \inf\{\ell \in \mathbb{R} : \mathbb{P}(L > \ell) \leq 1 - \alpha\},$$

for $a > 0$,

$$\begin{aligned}\text{VaR}_\alpha(aL + b) &= \inf\{\ell : \mathbb{P}(aL + b > \ell) \leq 1 - \alpha\} \\ &= \inf\left\{\ell : \mathbb{P}\left(L > \frac{\ell - b}{a}\right) \leq 1 - \alpha\right\}.\end{aligned}$$

Putting $\ell' = (\ell - b)/a$, so that $\ell = a\ell' + b$, gives

$$\begin{aligned}\text{VaR}_\alpha(aL + b) &= \inf\{a\ell' + b : \mathbb{P}(L > \ell') \leq 1 - \alpha\} \\ &= a \inf\{\ell' : \mathbb{P}(L > \ell') \leq 1 - \alpha\} + b \\ &= a \text{VaR}_\alpha(L) + b.\end{aligned}$$

2. For

$$X \sim N(\mu_1, \sigma_1^2), \quad Y \sim N(\mu_2, \sigma_2^2), \quad X \perp\!\!\!\perp Y,$$

we have

$$X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Let $z_\alpha = \Phi^{-1}(\alpha)$. For $\alpha \geq \frac{1}{2}$, $z_\alpha \geq 0$, and therefore

$$\begin{aligned}\text{VaR}_\alpha(X + Y) &= \mu_1 + \mu_2 + \sqrt{\sigma_1^2 + \sigma_2^2} z_\alpha \\ &\leq \mu_1 + \mu_2 + (\sigma_1 + \sigma_2) z_\alpha \\ &= \text{VaR}_\alpha(X) + \text{VaR}_\alpha(Y).\end{aligned}$$

3. Here $\lambda = 5$, $S_0 = 100$, and $X_1 = \log(S_1/S_0) \sim N(0, 0.1^2)$. Since $S_1 = S_0 e^{X_1}$,

$$L = -\lambda(S_1 - S_0) = -\lambda S_0(e^{X_1} - 1) = -500(e^{X_1} - 1).$$

Thus

$$f(x) = -500(e^x - 1).$$

Since $e^{X_1} \in (0, \infty)$,

$$L \in (-\infty, 500).$$

Let $q = \text{VaR}_\alpha(L)$. Then

$$\begin{aligned}\mathbb{P}(L \leq q) = \alpha &\iff \mathbb{P}(-500(e^{X_1} - 1) \leq q) = \alpha \\ &\iff \mathbb{P}\left(X_1 \geq \log\left(1 - \frac{q}{500}\right)\right) = \alpha.\end{aligned}$$

Hence

$$\log\left(1 - \frac{q}{500}\right) = F_X^{-1}(1 - \alpha).$$

For $\alpha = 0.99$,

$$F_X^{-1}(0.01) = -F_X^{-1}(0.99) \approx -0.23,$$

so

$$\text{VaR}_{0.99}(L) = 500(1 - e^{-0.23}) \approx 102.73.$$

4. For $T = 100$,

$$\frac{S_T}{S_0} = \prod_{t=1}^T \frac{S_t}{S_{t-1}} = \exp \left(\sum_{t=1}^T X_t \right).$$

Since $X_t \stackrel{\text{iid}}{\sim} N(0, 0.1^2)$,

$$Z := \sum_{t=1}^{100} X_t \sim N(0, 1), \quad L_{100} = -500(e^Z - 1).$$

Therefore

$$\text{VaR}_{0.99}(L_{100}) = 500 \left(1 - e^{-\Phi^{-1}(0.99)} \right) \approx 451.2.$$

Using $e^x - 1 \approx x$,

$$L_{100}^\Delta = -500Z.$$

Since Z and $-Z$ have the same distribution,

$$\text{VaR}_{0.99}(L_{100}^\Delta) = 500\Phi^{-1}(0.99) \approx 1163.2.$$

The linear approximation is poor over 100 days: it even exceeds the maximum possible loss of 500.

Seminar 4

1. For

$$\mathbb{P}(L \leq \ell) = 1 - \ell^{-2}, \quad \ell \geq 1,$$

let $q_u = \text{VaR}_u(L)$. Solving

$$1 - q_u^{-2} = u$$

gives

$$q_u = (1 - u)^{-1/2}.$$

Hence

$$\begin{aligned} \text{ES}_{0.99}(L) &= \frac{1}{1 - 0.99} \int_{0.99}^1 \text{VaR}_u(L) \, du \\ &= 100 \int_{0.99}^1 (1 - u)^{-1/2} \, du \\ &= 100 \left[-2(1 - u)^{1/2} \right]_{0.99}^1 \\ &= 200\sqrt{0.01}. \end{aligned}$$

Therefore

$$\text{ES}_{0.99}(L) = 20.$$

2. With total capital 300, the one-day linearised loss is

$$L^\Delta = -300 \sum_{i=1}^3 w_i X_i.$$

For

$$w_A = \left(\frac{1}{2}, \frac{1}{2}, 0\right), \quad w_B = \left(0, \frac{1}{2}, \frac{1}{2}\right), \quad w_C = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right),$$

we get

$$L_A^\Delta = -150(X_1 + X_2), \quad L_B^\Delta = -150(X_2 + X_3), \quad L_C^\Delta = -100(X_1 + X_2 + X_3).$$

Using

$$\Sigma = 10^{-4} \begin{pmatrix} 1 & 0.5 & 0 \\ 0.5 & 1 & -0.5 \\ 0 & -0.5 & 1 \end{pmatrix},$$

$$\text{Var}(X_1 + X_2) = 10^{-4} + 10^{-4} + 2(0.5 \cdot 10^{-4}) = 3 \cdot 10^{-4},$$

$$\text{Var}(X_2 + X_3) = 10^{-4} + 10^{-4} + 2(-0.5 \cdot 10^{-4}) = 10^{-4},$$

$$\begin{aligned} \text{Var}(X_1 + X_2 + X_3) &= 3 \cdot 10^{-4} + 2(0.5 + 0 - 0.5) \cdot 10^{-4} \\ &= 3 \cdot 10^{-4}. \end{aligned}$$

Thus

$$\text{sd}(L_A^\Delta) = 1.5\sqrt{3}, \quad \text{sd}(L_B^\Delta) = 1.5, \quad \text{sd}(L_C^\Delta) = \sqrt{3}.$$

Because all three losses are centred normal,

$$\frac{\text{VaR}_{0.99}(L_B^\Delta)}{\text{VaR}_{0.99}(L_A^\Delta)} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}, \quad \frac{\text{VaR}_{0.99}(L_C^\Delta)}{\text{VaR}_{0.99}(L_A^\Delta)} = \frac{2}{3}.$$

Therefore A is the riskiest strategy and B is the least risky.

For 10 independent days,

$$L_{B,10}^{\Delta} = -150 \sum_{t=1}^{10} (X_{t,2} + X_{t,3}) \sim N(0, 22.5),$$

because

$$\text{Var}(L_{B,10}^{\Delta}) = 10(1.5)^2 = 22.5.$$

Hence

$$\text{VaR}_{0.99}(L_{B,10}^{\Delta}) = 1.5\sqrt{10} \Phi^{-1}(0.99) \approx 11.0.$$

The limit is

$$0.20(300) = 60,$$

and $11.0 < 60$, so strategy B satisfies the constraint.

Seminar 5

3. For

$$L = 100T_4,$$

put

$$q_\alpha = F_4^{-1}(\alpha), \quad z_\alpha = \Phi^{-1}(\alpha).$$

Under the Student- t model,

$$\text{VaR}_\alpha^{(t)} = 100q_\alpha, \quad \text{ES}_\alpha^{(t)} = 100 \frac{f_4(q_\alpha)}{1-\alpha} \frac{4 + q_\alpha^2}{3}.$$

Also,

$$\text{sd}(L) = 100 \sqrt{\frac{4}{4-2}} = 100\sqrt{2}.$$

The matched normal model therefore gives

$$\text{VaR}_\alpha^{(N)} = 100\sqrt{2} z_\alpha, \quad \text{ES}_\alpha^{(N)} = 100\sqrt{2} \frac{\phi(z_\alpha)}{1-\alpha}.$$

	$\alpha = 0.99$	$\alpha = 0.999$
True t_4 VaR	374.7	717.3
Normal VaR	329.0	437.0
VaR understatement	12.2%	39.1%
True t_4 ES	522.1	968.6
Normal ES	376.9	476.2
ES understatement	27.8%	50.8%

The percentage understatement is

$$100 \frac{R^{(t)} - R^{(N)}}{R^{(t)}}.$$

The error increases sharply deeper in the tail because the Student- t distribution is much heavier-tailed than the normal distribution.

The ES formula used above follows from

$$L = \mu + sT, \quad T \sim t_\nu, \quad q = F_\nu^{-1}(\alpha).$$

Then

$$\begin{aligned} \text{ES}_\alpha(L) &= \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)] \\ &= \mu + s \mathbb{E}[T \mid T \geq q] \\ &= \mu + \frac{s}{1-\alpha} \int_q^\infty x f_\nu(x) \, dx. \end{aligned}$$

Writing

$$f_\nu(x) = C_\nu \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2},$$

we have

$$\frac{d}{dx} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu-1)/2} = -\frac{\nu-1}{\nu} x \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}.$$

Therefore

$$\int_q^\infty x f_\nu(x) \, dx = \frac{\nu + q^2}{\nu - 1} f_\nu(q),$$

and

$$\text{ES}_\alpha(L) = \mu + s \frac{f_\nu(F_\nu^{-1}(\alpha))}{1 - \alpha} \frac{\nu + [F_\nu^{-1}(\alpha)]^2}{\nu - 1}.$$

Seminar 8

1. For

$$F_{X,Y}(x,y) = (1 + e^{-x} + e^{-y})^{-1},$$
$$F_X(x) = \lim_{y \rightarrow \infty} F_{X,Y}(x,y) = (1 + e^{-x})^{-1},$$

and similarly

$$F_Y(y) = (1 + e^{-y})^{-1}.$$

The inverse standard logistic CDF is

$$F_X^{-1}(u) = \log \frac{u}{1-u}, \quad e^{-F_X^{-1}(u)} = \frac{1-u}{u}.$$

Thus

$$\begin{aligned} C(u,v) &= F_{X,Y}(F_X^{-1}(u), F_Y^{-1}(v)) \\ &= \left(1 + \frac{1-u}{u} + \frac{1-v}{v}\right)^{-1} \\ &= \frac{uv}{u+v-uv}. \end{aligned}$$

2. For

$$F_\theta(x,y) = (1 + e^{-x} + e^{-y} + (1-\theta)e^{-x-y})^{-1},$$

letting $y \rightarrow \infty$ gives

$$F_X(x) = (1 + e^{-x})^{-1},$$

and similarly for Y .

When $\theta = 0$,

$$1 + e^{-x} + e^{-y} + e^{-x-y} = (1 + e^{-x})(1 + e^{-y}),$$

so

$$F_0(x,y) = F_X(x)F_Y(y).$$

Hence X and Y are independent.

Substituting

$$e^{-x} = \frac{1-u}{u}, \quad e^{-y} = \frac{1-v}{v},$$

and multiplying the denominator by uv ,

$$\begin{aligned} uv + v(1-u) + u(1-v) + (1-\theta)(1-u)(1-v) \\ = 1 - \theta(1-u)(1-v). \end{aligned}$$

Therefore

$$C_\theta(u,v) = \frac{uv}{1 - \theta(1-u)(1-v)}.$$

3. Since

$$V_1^X = 10,000e^X,$$
$$V_1^X < 9,000 \iff X < \log(0.9),$$

and similarly for Y .

For a Clayton copula,

$$\tau = \frac{\theta}{\theta + 2}.$$

Thus

$$\frac{\theta}{\theta + 2} = \frac{1}{3} \implies \theta = 1.$$

The marginal probabilities are

$$\begin{aligned} u &= \Phi\left(\frac{\log 0.9}{0.1}\right) = \Phi(-1.05361) \approx 0.146032, \\ v &= \Phi\left(\frac{\log 0.9}{0.06}\right) = \Phi(-1.75601) \approx 0.039543. \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{P}(V_1^X < 9,000, V_1^Y < 9,000) &= C_1(u, v) \\ &= (u^{-1} + v^{-1} - 1)^{-1} \\ &\approx 0.032117. \end{aligned}$$

Therefore

$$\mathbb{P}(V_1^X < 9,000, V_1^Y < 9,000) \approx 3.21\%.$$

Seminar 11

1. For the exponential distribution

$$F(x) = 1 - e^{-\beta x},$$

choose

$$c_n = \frac{1}{\beta}, \quad d_n = \frac{\log n}{\beta}.$$

Then

$$c_n x + d_n = \frac{x + \log n}{\beta},$$

and

$$F(c_n x + d_n) = 1 - e^{-(x + \log n)} = 1 - \frac{e^{-x}}{n}.$$

Therefore

$$F^n(c_n x + d_n) = \left(1 - \frac{e^{-x}}{n}\right)^n \longrightarrow \exp\{-e^{-x}\}.$$

Thus the limit is Gumbel, with

$$\xi = 0.$$

2. For the Pareto distribution

$$F(x) = 1 - \left(\frac{\kappa}{\kappa + x}\right)^\alpha,$$

choose

$$c_n = \frac{\kappa n^{1/\alpha}}{\alpha}, \quad d_n = \kappa n^{1/\alpha} - \kappa.$$

Then

$$\kappa + c_n x + d_n = \kappa n^{1/\alpha} \left(1 + \frac{x}{\alpha}\right),$$

so

$$F(c_n x + d_n) = 1 - \frac{1}{n} \left(1 + \frac{x}{\alpha}\right)^{-\alpha}.$$

For $1 + x/\alpha > 0$,

$$\begin{aligned} F^n(c_n x + d_n) &= \left[1 - \frac{1}{n} \left(1 + \frac{x}{\alpha}\right)^{-\alpha}\right]^n \\ &\longrightarrow \exp\left\{-\left(1 + \frac{x}{\alpha}\right)^{-\alpha}\right\}. \end{aligned}$$

Thus the limit is Fréchet, with

$$\xi = \frac{1}{\alpha} > 0.$$